

Changes in soil organic carbon in a clay loam soil following ploughing and reseedling of permanent grassland under temperate moist climatic conditions

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Abstract

This study investigated changes in soil organic carbon (SOC) in a poorly drained clay loam soil under (i) permanent grassland (PG) over 7 years and (ii) after PG renovation over 2.5 years. The experimental area was divided into four blocks with four paddocks per block. Composite soil samples from each PG block were taken to 30-cm depth on five occasions between 2004 and 2011. In June 2008, one paddock per block was renovated by ploughing and reseedling and sampled as above on the latter four occasions. Renovation decreased SOC ($P < 0.05$) by 32.2 t ha^{-1} in the 2.5 years following ploughing; the difference developed almost entirely (86%) in the first four months. Renovation had no effect on above-ground productivity, standing root and stubble phytomass or on dissolved organic matter leaching. Therefore, soil respiration was considered to be a single potential pathway responsible for the SOC loss from renovated grassland. Although a simple linear regression indicated a tendency for PG to accumulate SOC, there was no evidence of recovery in SOC to previous levels following renovation during the study period. These results could have implications for greenhouse gas inventories in countries where PG is an important land-use type.

Keywords: soil organic carbon, permanent grassland, ploughing, renovation, DNDC

Introduction

Grasslands represent an important part of the global carbon (C) cycle (Jones *et al.*, 2006) and act as an important C reservoir (Schimel *et al.*, 1995; Fontaine *et al.*, 2007). Worldwide, the surface soils of temperate grassland store around 295 GT of soil organic C (SOC) (Bolin and Sukumar, 2000). While it is well known that grassland ecosystems have a high SOC content and are a potential sink or source of C (Arshad *et al.*, 2004; Bronson *et al.*, 2004; Soussana *et al.*, 2007; Rogiers *et al.*, 2008), information on SOC storage and dynamics in grassland soils varies between studies (Post and Kwon, 2000; Leifeld *et al.*, 2005). Accumulation or depletion of SOC is predominantly dependent upon the ability of the soil to stabilize C, the physical protection of C within soil aggregates, the lability of C added to the soil, the balance of inputs (e.g. photosynthesis and organic matter inputs) and losses (e.g. ecosystem respiration, product exports, leaching and erosion) and also the management practices and climate variations that alter this balance (Saggar and Hedley, 2001; Freibauer *et al.*, 2004; Schipper *et al.*, 2010). Therefore, SOC accumulation is generally promoted by (i) high clay content (Wang *et al.*, 2003; Jones and Donnelly, 2004; Mestdagh *et al.*, 2006), which physically stabilizes soil organic matter (SOM) on clay surfaces (Post *et al.*, 2001; Guggenberger and Kaiser, 2003; Lal, 2004; Swanston *et al.*, 2005; von Lutzow *et al.*, 2006); (ii) high soil moisture (Jones *et al.*, 2006; Mestdagh *et al.*, 2006; Hutchinson *et al.*, 2007; O'Brien *et al.*, 2010) and intermittent waterlogging, which impede decomposition processes

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(Davidson and Lefebvre, 1993); (iii) low temperatures which also lead to lower decomposition rates (Freibauer *et al.*, 2004; Hutchinson *et al.*, 2007); (iv) increased intensity and pattern of precipitation events (Springob and Kirchmann, 2004; O'Brien *et al.*, 2010); (v) high primary production and thus high C inputs to the soil (Campbell *et al.*, 2001; Jones and Donnelly, 2004); and (vi) high groundwater tables promoting higher phytomass production by easing moisture limitation during summer (Vertes *et al.*, 2004; Phelan *et al.*, 2012).

The yields and forage quality of temperate managed permanent grassland (PG) progressively deteriorate due to the loss of the original sown sward components and their replacement by unsown species, due to sward compaction caused by inappropriate management or occasional sward damage due to extreme weather events (Humphreys and Lawless, 2008; Creighton *et al.*, 2010; Velthof *et al.*, 2010). For these reasons, it is common practice for grassland swards to be renovated by ploughing and reseeded, typically at intervals of 5–10 years (Humphreys *et al.*, 2009a). Ploughing of grassland leads to physical disruption of the soil structure that protects SOC, increased aeration and, consequently, rapid microbial decomposition of SOC (Six *et al.*, 2004; Buss *et al.*, 2005; Velthof *et al.*, 2010). The disruption also decreases soil moisture, increases maximum soil temperature and thus accelerates the mineralization process (Davidson *et al.*, 2000; Lal, 2004; Angers *et al.*, 2010). For these reasons, occasional renovation of PG, particularly of long-term grassland with a substantial SOC pool, may temporarily lead to significant carbon dioxide (CO₂) emissions. Rising concentrations of greenhouse gases (GHG) in the atmosphere are of worldwide concern, which is driven by the need for industrialized countries to reduce their GHG emissions to meet stringent targets established by the Kyoto Protocol of the United Nations Framework Convention on Climate Change. Furthermore, the signatories of the Kyoto Protocol are also committed to establish national inventories of SOC content and its changes. The inventory systems are being developed but, to date, national-scale measurement-based inventories have not been able to clearly attribute the changes in SOC to specific land-use or climatic effects (Smith *et al.*, 2007). There is therefore a strong need for accurately quantifying the effects of different management practices on SOC content.

The magnitude of change in SOC following a change in land use after a certain time period is generally quantified as the differences in SOC content between that of plots under new land use and that of plots under unchanged management (Ellert *et al.*, 2001; Schipper *et al.*, 2010). So far, the impact of land

use or management on SOC has focused on changes following afforestation of grassland (Guo and Gifford, 2002; Laganière *et al.*, 2010), conversion of grassland to cropland (Johnston, 1986; Strebel *et al.*, 1988; Guo and Gifford, 2002; Lal, 2002; Ogle *et al.*, 2005), conversion of forest to grassland or cropland (Murty *et al.*, 2002) and management of cropland (Senthilkumar *et al.*, 2009). Grassland management practices such as ploughing and reseeded of long-term PG might also lead to important SOC change, contribute to CO₂ emissions and thus significantly affect the GHG budget. This could be particularly important in countries like Ireland, where over 90% of the agricultural land is classified as permanent pasture (Donnelly *et al.*, 2011) and the predominant land use is production of perennial forage. In the context of climate change, it is therefore important to understand how grassland renovation influences SOC losses.

The objectives of the work were to quantify changes in SOC content on a poorly drained clay loam soil under (i) PG used for dairy farming in southern Ireland over 7 years and (ii) after PG renovation over two-and-a-half years. The grassland renovation in this context refers to ploughing followed by reseeded. To explain which factors have a major influence on SOC dynamics, most of the C inputs and C losses through leaching in permanent and renovated grassland (RG) were described and quantified. Field observations of change in SOC content were compared with simulations of SOC dynamics using the denitrification–decomposition (DNDC) model.

Materials and methods

Site description

The study was conducted on a dairy research farm in Solohead, Tipperary, Ireland (52°51'N, 08°21'W; 150 m a.s.l.) from October 2004 to January 2011. The local climate is humid temperate oceanic, and there is a long potential growing season from February to mid-December. The farm has been under PG used primarily for grazing, and to a lesser extent for silage production, for more than 50 years. During the last 10 years, the mean growing season length, based on a soil temperature above 5°C at 5-cm depth, was approximately 305 days, and the annual grass production rates, supported by slurry and N fertilizer (Table 2), ranged from 13.5 to 14.7 t dry matter (DM) ha⁻¹.

The predominant soils are poorly drained gleys (90%) and grey-brown podzolics (10%) with a clay loam texture and low permeability (28% clay, 35% silt). The soils have a high buffering capacity with cation exchange capacity of 25.4 cmol kg⁻¹. The mean SOC and total nitrogen (N) content at 0- to 30-cm

depths is 4.48 and 0.48 mg kg⁻¹, respectively, with a corresponding soil bulk density of 1.09 g cm⁻³. Soil depth to bedrock (Cappagh White Devonian Sandstone Formation (Archer *et al.*, 1996)) is uneven, ranging from 5 to 10 m. Overlying quaternary till contains a perched water table (depth of 0–2.2 m below ground level (bgl)). Although a number of ditches (4 m bgl) and tile drains (1.8 m bgl with spacing of 25 m) were installed between 1960 and 1995 across the farm to improve the drainage on site by lowering the water table (Tim Gleeson, personal communication), much of the farm still remains seasonally wet, waterlogged or flooded due to impeded drainage.

Mean annual rainfall, soil moisture deficit and soil temperature at 10-cm depth for the period from January 2004 to December 2010 were 1055 mm, 10.7 mm d⁻¹ and 9.8°C respectively. While the annual rainfall in 2008 and 2009 was exceptionally high and exceeded the 10-year mean by 21% and 27%, respectively, the annual rainfall in 2010 was only 857 mm (Table 1). Uncharacteristically wet weather conditions in 2008 and 2009 were also evident from water-filled pore space records at 5-cm depth indicating complete soil saturation (≥ 1) for 14 months.

Experimental design and treatments

During 1999, the area of the farm (52 ha) was divided into eight blocks (>6 ha). Four of these blocks were assigned to this experimental study (24 ha). Each block was divided into four paddocks (>1 ha). All the paddocks had been renovated between 1985 and 1995. From 2004 to 2011, the farm was under typical grassland management as described in Table 2. The pasture was dominated by perennial ryegrass (*Lolium perenne* L.) with white clover (*Trifolium repens* L.) and was allocated to livestock in a rotational stocking system. The RG treatment was established in June 2008, when one paddock in each block was grazed or cut to a height of 5 cm, inversion-ploughed to a depth of approximately 20 cm and sown with a seed mixture containing perennial grass (cv. Aran) and white clover (cv. Susi).

Soil organic carbon

Soil samples from the PG were taken in October 2004, October 2008, February 2009, December 2009 and January 2011. The RG was sampled on the latter four dates. At each sampling, fifteen cores per paddock were taken randomly using a Geonor MCL3 hydraulic auger (Geonor A/S, Oslo, Norway) to a depth of 30 cm. Samples were bulked to provide a composite sample per block (i.e. one per replicate). Immediately

after sampling, all plant material and stones were removed manually and were omitted from the calculation of SOC. The soil cores were oven-dried at 40°C, crushed, sieved (2 mm) and analysed for total C, inorganic C and total N content by dry combustion at 1300°C using a LECO CN-2000 elemental analyzer (Leco, St. Joseph, MI, USA). Subsamples analysed for inorganic C were pre-treated by heating at 450°C, and SOC concentration was then estimated by the difference between total C and inorganic C.

Soil bulk density

Soil bulk density was measured in August 2009 using the cylinder core method (Blake and Hartge, 1986). Four replicate samples were taken at six locations from both PG and RG. A steel cylinder (internal $\varnothing = 4.6$ cm; $L = 5$ cm) was hammered into the soil sufficiently to fill the inner cylinder without any soil compression at the required depth. Soil cores obtained by emptying the cylinders were examined for signs of shattering or compression. On return to the laboratory, the samples were dried at 105°C and weighed prior to bulk density determination.

Soil organic C content to a 30-cm depth was calculated on an equivalent mass basis as a product of SOC concentration, bulk density and the layer depth as described by Ellert *et al.* (2007):

$$\text{SOC (t C ha}^{-1}\text{)} = \text{SOC concentration (mg C g}^{-1}\text{ dry soil)} \times \text{bulk density (g cm}^{-3}\text{)} \times \text{sampling depth (cm)}.$$

Root, stubble and herbage phytomass

Root standing phytomass was sampled using a Dutch hand auger in September 2010. Four cores per depth section were obtained for PG at six locations and for RG at three locations. Prior to taking the soil cores, the herbage within a square of 50 cm² was cut using Gardena hand shears (Gardena, Accushears, Ulm, Germany) to a height of 5 cm. Soil cores ($\varnothing = 6$ cm) were taken at depths of 0–15 and 15–30 cm. Following storage in a refrigerator (4°C) overnight, the cores were soaked in tap water for approximately 30 min. Roots were separated from the soil by washing with running water through a 1-mm sieve and were picked out with forceps. The washed roots (>1 mm) were oven-dried at 40°C for 24 h prior to DM determination by weighing. Hence, roots in the current context refer to a root material extracted as described above, which included dead roots. The stubble phytomass (0–5 cm) was measured at the same locations; samples were taken from each of the topsoil cores by harvesting the herbage to ground level using scissors. The DM was determined as above. During the grazing

Table 1 Weather data at Solohead Research Farm from 2004 to 2010

	2004	2005	2006	2007	2008	2009	2010	10-year mean
Annual rainfall (mm)	1000	885	1121	996	1446	1403	857	1017
Highest monthly rainfall (mm)	186 (October)	128 (October)	178 (September)	138 (December)	213 (January)	221 (November)	123 (September)	114 (October)
Lowest monthly rainfall (mm)	37 (February)	32 (June)	20 (June)	13 (April)	37 (May)	19 (February)	34 (August)	60 (April)
Effective rainfall (mm)	551	394	644	419	900	866	346	552
Drainage season	January–March August–December	January–March October– December	January–March September– December	January–March December	January–March June–December	January–April July–December	January–March September– December	January–March October–December
Mean soil moisture deficit (mm d ⁻¹)	9.2	12.1	16.0	11.2	4.4	5.9	15.9	10.5
Mean soil temperature (°C)	10.1	10.5	10.0	10.2	9.5	9.6	8.9	10.6
Highest monthly temperature (°C)	16.1 (August)	16.8 (July)	17.1 (July)	15.6 (August)	15.7 (August)	15.7 (July)	15.9 (July)	16.2 (July)
Lowest monthly temperature (°C)	4.6 (January)	5.0 (February)	4.3 (February)	5.1 (February)	5.1 (December)	4.1 (January, December)	2.5 (February, December)	5.2 (February)

Table 2 Grassland management at Solohead Research Farm from 2004 to 2009

Treatments		Livestock density LU ha ⁻¹	Grazing days per year d yr ⁻¹	Annual herbage intake by livestock t ha ⁻¹	Mean white clover content in the swards %	Annual N input through biological N fixation* kg N ha ⁻¹	Annual N input in fertilizer kg N ha ⁻¹	Annual slurry application kg N ha ⁻¹
2004†	A	2.2	254	12.6	6.7	14	224	92
	B	2.2	254	11.9	32.1	129	84	92
2005†	A	2.2	256	12.9	4.5	10	219	84
	B	2.2	250	12.0	28.2	113	89	86
2006†	A	2.2	255	11.5	6.6	11	240	85
	B	2.2	251	10.2	26.6	87	99	79
2007‡	C	2.1	232	10.5	20.1	60	90	
2008§	D	2.1	220	10.2	22.0	73	105	138
	E	1.6	234	8.6	28.2	111	0	115
2009§	F	1.5	231	9.3	20.6	99	0	87
	D	2.1	218	9.2	19.6	60	96	80
	E	1.6	234	8.2	30.6	114	0	103
	F	1.5	235	9.5	35.4	169	0	93

*Estimated according to an empirical model based on herbage yield and white clover content (Humphreys *et al.*, 2008).

†Humphreys *et al.*, 2009b. ‡Humphreys *et al.*, 2010. §Humphreys *et al.*, 2010; Keogh *et al.*, 2010.

period, pre-grazing herbage in each paddock was measured by cutting four random strips (5 × 0.55 m) using a HRH-536 lawn-mower (Honda, Georgia, USA) at a cutting height of 5 cm. The cut herbage was bulked and weighed, and DM of a subsample was determined as described above. The above-ground phytomass productivity was obtained as a sum of pre-grazing herbage masses from the whole grazing season and stubble phytomass. The phytomass subsamples taken for elemental analyses were milled through a 0.5-mm sieve before being analysed for total C content by dry combustion at 1050°C using a LECO CN-2000 elemental analyzer. The C stored in phytomass was estimated from its C content.

Dissolved organic carbon in groundwater

Dissolved organic-C leaching was investigated through sampling of shallow groundwater (0–2.2 m bgl) as described by Necpálová *et al.* (2012). In July 2008, ten piezometers, constructed from PVC pipes, were installed in each paddock. The slotted screen opening interval on the lower 20 cm of the casing was covered by a filter sock. The annulus (<3 cm) between the casing and the piezometer wall was grouted with sand and bentonite on the soil surface to prevent water channelling down the outside of the casing. Samples were taken at 14-day intervals, after purging the piezometers, during the main winter-drainage period and following periods of high rainfall during other times of the year. Dissolved organic N in the samples was

determined from August 2008, but, for organizational reasons, the DOC was determined for the first time in March 2009 and then on another eight occasions until January 2010. On each occasion, a 50-ml sample was taken from each piezometer and bulked into two composite samples per paddock. All samples were filtered (Whatman no.2) within an hour after sampling and then analysed for DOC by chemiluminescence after catalytic combustion at 720°C using a Shimadzu TOC-VCPH analyzer with an ASI-V autosampler. Estimated vertical travel time through the unsaturated zone to groundwater ranged between 0.21 and 0.57 years at different locations around the farm (Necpálová *et al.*, 2012). The losses from each treatment were estimated by multiplying the mean concentrations recorded on the sampling date with the volume of effective rainfall (ER, drainage) between two sampling occasions, with an assumption that the DOC concentrations recorded in the groundwater were representative of concentrations in the ER reaching the piezometer screen.

Statistical analyses

Statistical analyses were performed using the SAS software, version 9.01 (SAS Institute Inc., Cary, NC, USA). Normality distribution of residuals was tested. Transformation of variables was not required in any case. Soil organic C and C/N ratio were analysed using a two-way analysis of variance. The time factor such as sampling date was entered as a repeated measure with a compound symmetry covariance structure, and

the other factors (renovation and renovation $\times$ sampling date interaction) were entered as fixed effects. Similarly, DOC concentrations in groundwater between PG and RG were subjected to a repeated-measure ANOVA model. A one-way analysis of variance was used to examine the differences in above-ground productivity, standing root phytomass and DOC annual losses between PG and RG. *Post hoc* treatment comparisons were made according to Tukey's and Bonferroni tests. Simple linear regression analysis was used to estimate the annual change in SOC under PG as a function of time. In addition, the differences in SOC under PG and associated C/N ratio over time were also examined using a one-way analysis of variance.

DNDC modelling

The dynamics of SOC were simulated using the DNDC model (version 9.5). The DNDC is a process-based computer simulation model of soil C and N biogeochemistry in agroecosystems (Li *et al.*, 1992a,b, 1994, 2000; Li, 1996). It operates at a daily time step and consists of two components: the first component consisting of the soil climate, crop growth and decomposition submodels, and the second component, consisting of the nitrification, denitrification and fermentation submodels. The entire model forms a bridge between C and N biogeochemical cycles and the basic ecological drivers.

In DNDC, SOC resides in four major pools: plant residue, microbial biomass, humads (i.e. active humus) and passive humus. Each pool consists of two or three subpools with different specific decomposition rates. The daily decomposition rate for each subpool is regulated by the pool size, the specific decomposition rate, soil clay content, N availability, soil temperature and soil moisture. When SOC in a pool decomposes, the decomposed C is partially allocated to other SOC pools and partially lost as CO₂. Dissolved organic C is produced as an intermediate during decomposition and can be immediately consumed by the soil microbes (Byrne and Kiely, 2009).

The SOC dynamic was simulated for one PG and one RG paddock in each block, i.e., eight paddocks in total. The models were parameterized using paddock-specific soil properties (bulk density, pH, texture, SOC, soil ammonium and nitrate concentration) and driven using climatic (i.e. daily rainfall, daily minimum and maximum temperature, N concentration in rainfall) and management data (i.e. slurry and fertilizer applications, grazing, cutting for silage, ploughing) over a 2-year period. The management schedule was consistent with the management operations carried out in selected paddocks in 2008 and 2009. Field capacity,

wilting point and saturated hydraulic conductivity were calculated based on the measured soil properties using the soil water characteristics software, version 6.02.74 (USDA Agricultural Research Service, Washington DC). To stabilize the crop growth submodel and reduce the residual effects of initial conditions, we extended the simulation period to seven consecutive years, discarding the first 5 years of each simulation. The models were calibrated using annual grass production from 2008 and 2009, and the calibration was validated against daily soil moisture, soil temperature data and SOC from February 2009. The same model simulations were conducted for calibration and validation. The fit between simulated and observed values over the course of simulation was assessed using correlation coefficient (R^2).

Results

Effect of grassland renovation on bulk density and SOC content

Mean bulk density ($\pm$ standard error, SE) measured 15 months following renovation in PG and RG topsoil (0–30 cm) was 1.09 ± 0.03 g cm⁻³ (coefficient of variation; CV = 15%) and 1.02 ± 0.04 g cm⁻³ (CV = 21%), respectively, and was not affected by grassland renovation ($P = 0.4$). The RG treatment was established in June 2008; however, as SOC content had not been measured immediately before ploughing, the content from October 2004 was considered as a reference value for the comparisons (Figure 1). Renovation decreased topsoil SOC ($P < 0.05$; Figure 1) resulting in means of 146.81 ± 16.98 and 114.61 ± 13.51 t SOC ha⁻¹ in PG

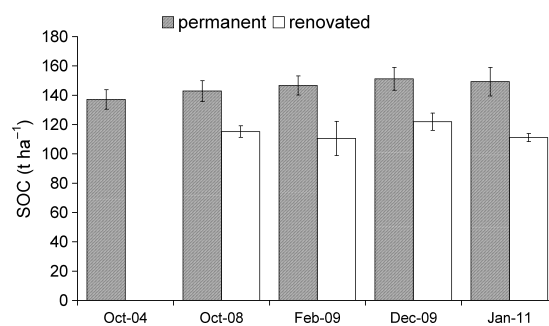


Figure 1 Soil organic carbon (SOC) at 0- to 30-cm depths under permanent and renovated grassland on four sampling occasions after grassland renovation compared with SOC levels from October 2004. Error bars are $\pm$ SE. Means across the treatments are significantly different on the basis of analysis of variance ($P = 0.005$) and the Tukey's test; means across sampling dates are not significantly different ($P > 0.05$).

and RG respectively. The mean difference between PG and RG represented a loss of 32.2 t C ha^{-1} , which is a 22% reduction in topsoil SOC compared with PG in the 2.5 years following renovation (June 2008 to January 2011). This difference developed almost entirely during the first four months following renovation (June to October 2008). As the renovation had no effect on bulk density, the difference in SOC between the treatments resulted from a difference in SOC concentrations. The C/N ratio ranged from 8.9 to 9.8 and was not affected by renovation (Table 4).

Effect of grassland renovation on above-ground and below-ground phytomass

The C content of herbage was 434 mg g^{-1} . The annual herbage production obtained as a sum of continuously measured pre-grazing herbage masses was 9.1 ± 0.29 and $9.6 \pm 0.55 \text{ t DM ha}^{-1}$ in PG and RG, respectively, and did not differ between the treatments in the 2 years following renovation (2008 and 2009).

The stubble phytomass (<5 cm) of PG and RG amounted to 2.2 ± 0.53 and $2.6 \pm 0.33 \text{ t DM ha}^{-1}$. Its C contents for PG and RG were 402 and 385 mg g^{-1} , respectively, and consequently, the stubble stored $0.9 \pm 0.21 \text{ t C ha}^{-1}$ in PG and $1.0 \pm 0.13 \text{ t C ha}^{-1}$ in RG (Table 3). The annual above-ground productivity of PG was $11.2 \text{ t DM ha yr}^{-1}$ ($4.8 \text{ t C ha}^{-1} \text{ yr}^{-1}$). This value was not significantly different from the $12.2 \text{ t DM ha yr}^{-1}$ ($5.2 \text{ t C ha}^{-1} \text{ yr}^{-1}$, Table 3) in RG.

The root standing phytomass sampled at 27 months following renovation in the upper 30 cm of the soil profile under PG and RG amounted to 16.0 and $14.9 \text{ t DM ha}^{-1}$, respectively, and there were no differences between the treatments (Table 3). The major part of the root system under both PG and RG (91%

and 96%) was located in the upper 15 cm of the soil profile and decreased rapidly at 15- to 30-cm depths. The C content of roots in PG at 0–15 cm and at 15–30 cm was 255 and 356 mg g^{-1} respectively. These values were similar to the 282 and 361 mg g^{-1} in RG. The C content of roots was low compared with C content of herbage and stubble. Despite this, the root systems under PG and RG stored 3.9 ± 1.06 and $4.1 \pm 1.06 \text{ t C ha}^{-1}$ at 0 to 15 cm and 0.3 ± 0.07 and $0.1 \pm 0.04 \text{ t C ha}^{-1}$ at 15 to 30 cm respectively (Table 3). The roots were the dominant soil C input in both of the grassland treatments. As the root phytomass was measured only on a single occasion, the annual below-ground productivity, and therefore the annual root C inputs to soil via root mortality, could not be estimated on the basis of the available data.

Overall, the renovation had no effect on phytomass sampled 27 months following ploughing, its C storage and associated C/N ratio, and it also had no effect on above-ground phytomass productivity during the two production years following renovation (Table 3).

Effect of grassland renovation on DOC leaching to groundwater

The rainfall and ER values for the sampling period from March 2009 to January 2010 were 716 and 545 mm respectively. There was a two-way interaction ($P < 0.001$) between the effect of renovation and sampling date on DOC concentration and its corresponding C/N ratio. The DOC under PG ranged from 0.2 to 70.9 mg L^{-1} (mean = 13.6 ± 0.61) with corresponding mean C/N ratio of 12.2 ± 0.72 . The DOC under RG was recorded within the range of 0.13 to 67.6 mg L^{-1} (mean = 10.7 ± 0.78) with a mean C/N ratio of 15.7 ± 1.7 . The annual DOC leaching losses to groundwater from PG and RG were not different and

Table 3 Above-ground and below-ground phytomass and dissolved organic carbon (DOC) losses to groundwater from permanent (PG) and renovated grassland (RG) at Solohead Research Farm in 2008 and 2009

	PG		RG		Significance	PG		RG		Significance
	Dry mass t ha ⁻¹	SE	Dry mass t ha ⁻¹	SE		C t ha ⁻¹	C:N	C t ha ⁻¹	C:N	
Above-ground primary productivity:										
Utilized phytomass > 5 cm	9.1	0.32	9.6	0.38	<i>P</i> > 0.05	3.9	16.50	4.2	16.50	<i>P</i> > 0.05
Stubble < 5 cm	2.2	0.53	2.6	0.33	<i>P</i> > 0.05	0.9	18.63	1.0	20.70	<i>P</i> > 0.05
Below-ground standing root phytomass:										
0–15 cm	15.2	4.15	14.6	3.77	<i>P</i> > 0.05	3.9	16.16	4.1	19.45	<i>P</i> > 0.05
15–30 cm	0.8	0.19	0.3	0.10	<i>P</i> > 0.05	0.3	25.70	0.1	29.59	<i>P</i> > 0.05
0–30 cm	16.0		14.9			4.2		4.2		
DOC leaching						0.1	12.20	0.1	15.70	<i>P</i> > 0.05

were estimated at 74.0 ± 5.6 and 62.4 ± 6.4 kg ha⁻¹ respectively (Table 3).

SOC change over time under permanent grassland

The slope of linear regression line reflects the rate of SOC accumulation/depletion on a horizontal land area basis. In this case, the slope indicated a tendency for SOC to increase over time, although this increase was not significant ($R^2 = 0.07$, $P > 0.05$; Figure 2). The sampling date had no effect on SOC content over the entire study period. Also, C/N ratio remained stable around 9.3 ± 0.05 without significant variation between the years (Table 4).

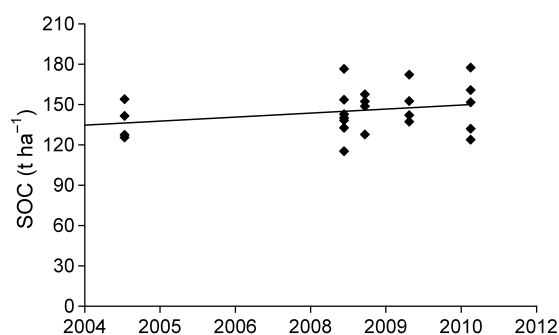


Figure 2 Soil organic carbon (SOC) change under permanent grassland at 0 to 30 cm from 2004 to 2011; the line was fitted using the linear regression equation: $y = 2.13x + 136.5$, where x represents the year of successive sampling, $R^2 = 0.0692$, $P > 0.05$. Each symbol represents a single result for a composite soil sample.

Table 4 C/N ratio (mean and standard deviation, SD) of soil organic matter at 0- to 30-cm depths under permanent and renovated grassland on four sampling occasions after grassland renovation compared with C/N ratio from October 2004. Means are not significantly different on the basis of analysis of variance ($P > 0.05$)

Date	Permanent grassland		Renovated grassland	
	C:N ratio	SD	C:N ratio	SD
October 2004	9.21	0.11		
October 2008	9.32	0.12	8.97	0.13
February 2009	9.37	0.16	9.30	0.14
December 2009	9.41	0.13	9.41	0.17
January 2011	9.28	0.12	9.19	0.18

DNDC modelling

The calibration of DNDC resulted in a good statistical fit between model outputs and the Solohead data for grass production in 2008 and 2009 ($R^2 = 0.76$, $P < 0.05$) and SOC in February 2009 ($R^2 = 0.98$, $P < 0.001$). The simulated daily soil moisture (for February, May, July and September, $R^2 = 0.77$ – 0.88 , $P < 0.001$; other months, $R^2 = 0.47$ – 0.65 , $P < 0.001$, $n = 30$) and soil temperature ($R^2 = 0.92$, $P < 0.001$, $n = 365$) significantly correlated with the observed values. The DNDC simulations of PG predicted an annual increase in SOC at 0- to 10-cm (0.28 ± 0.087 t C ha⁻¹) and decrease at 10- to 20-cm (0.11 ± 0.005 t C ha⁻¹) and at 20- to 30-cm (0.07 ± 0.002 t C ha⁻¹) depths over the course of simulation. When summed for the upper 30 cm of the soil profile, the simulations predicted SOC accumulation in PG at a rate of 0.10 t C ha⁻¹ yr⁻¹ (i.e. a 0.07% increase). Following the renovation (ploughing to 20-cm depth), the simulated SOC content in RG decreased annually by 4.44 ± 0.285 t C ha⁻¹ at 0–10 cm and by 1.15 ± 0.089 t C ha⁻¹ at 10- to 20-cm depths. Its content, however, increased by 2.77 ± 0.144 t C ha⁻¹ yr⁻¹ at 20- to 30-cm depths. Overall, the simulation of grassland renovation situation predicted a loss of 2.82 t C ha⁻¹ yr⁻¹ from the topsoil (i.e. a 1.97% reduction).

Discussion

Effect of grassland renovation on C inputs to SOC pool

In grazed grasslands, soil C inputs enter the SOC pool in the form of above-ground litter, root phytomass or animal excreta (Bol *et al.*, 2004). Due to the perennial nature of the plants and high below-ground growth allocation, the levels of C inputs under PG are generally high (Loiseau and Sousana, 1999). Grassland renovation causes death of most of the plants that form this ecosystem (Vertes *et al.*, 2007) and at the same time an induced mineralization of phytomass residues, and labile SOM increases herbage yields in the subsequent years. The absence of an increase in the above-ground phytomass productivity of RG in the 2 years following renovation in the current study may have resulted from a combination of factors limiting herbage production (i.e. low soil temperature and high rainfall in 2008 and 2009) and increasing susceptibility to treading damage during these years (Phelan *et al.*, 2012). The main input of C to the soil in both grassland types, PG and RG, came from roots (exudates and dead roots). As the root systems are subjected to simultaneous growth, senescence and decay,

an accurate determination of annual below-ground productivity has proven to be difficult (Aerts *et al.*, 1989; Gill and Jackson, 2000; Whitehead, 2000; Milchunas and Lauenroth, 2001), especially with grazing animals present (Vertes *et al.*, 2007). The focus of the current study was to compare the root phytomass in PG and RG and thus the potential C inputs through rhizodeposition. The comparison was therefore based only on a single-core sampling occasion and was representative of a fine-root standing phytomass grown 27 months following renovation. The absence of any difference between PG and RG in terms of root phytomass and its associated C content suggested that the root systems in RG developed and accumulated fast and provided root phytomass that was comparable with that of PG after 27 months. This is contrary to the findings of Whitehead (2000) who reported that newly sown grass sward produces about 10 t DM ha⁻¹ after 3 or 4 years, and this is still not comparable with the root phytomass of PG (approximately 15 t ha⁻¹). The root growth and branching following reseeding in the current study was likely to have been promoted by frequent regrowth following grazing events and by the effects of the clay loam texture protecting root systems from moisture- and nutrient-access limitation during summer months (Robinson, 1994). The rotational stocking and slurry management practised on the farm resulted in an equal amount of deposited excreta and slurry across the treatments. As a result, overall, there was no difference in the quantity and quality of soil C inputs between the treatments 27 months following grassland renovation.

Effect of grassland renovation on SOC content

The losses of SOC supported the hypothesis that grassland renovation causes a strong and sudden decrease in SOC in the soil profile. It has been reported that this decrease can be attributed to reduced soil C inputs, increased decomposability of phytomass residues or cultivation effects that decrease the amount of physical protection (Post and Kwon, 2000). As the measured C inputs did not differ between the treatments, the decrease in SOC content following grassland renovation was likely to have been a result of soil disturbance by ploughing.

It is generally accepted that disruption of soil aggregates increases soil aeration, which exposes stable, adsorbed SOM within micro- and macro-aggregates to rapid microbial decomposition and thus causes substantial losses (Elliot, 1986; Augustin *et al.*, 1996; Kasimir-Klemedtsson *et al.*, 1997; Reicosky *et al.*, 1999; Six *et al.*, 2002; Vertes *et al.*, 2004; Lenhart *et al.*, 2007). A number of studies have published results showing rapid losses of SOC following conversion of temperate

grassland to cultivated use (Tiessen *et al.*, 1982; Mann, 1986; Burke *et al.*, 1989; Schlesinger and Andrews, 2000; Slobodian *et al.*, 2002; Schnitzer *et al.*, 2006). For example, Strebel *et al.* (1988) measured losses of up to 100 t ha⁻¹ of SOC (57%) in 4 years following conversion of PG to arable land. However, there have been only a few studies on the effect of renovation of grassland on SOC dynamics; for example, Eriksen and Jensen (2001) measured soil respiration directly using a dynamic-chamber method during the first three months following ploughing of an old grass-clover pasture in grass-arable rotation on a sandy soil (57% sand). Their results showed that cultivation caused losses of 1.2 t C ha⁻¹ over a three-month period, compared with undisturbed grassland. The magnitude of losses in the current study was several-fold higher. This could be attributed to the fact that the PG on this site was based on a heavy clay loam soil and had been accumulating SOC in the soil profile for more than 50 years under unchanged management. However, the fact that the difference in SOC developed almost entirely (86%) in the first four months following ploughing is in agreement with previous cultivation studies (e.g. Vertes *et al.*, 2007).

The DOC losses measured after March 2009 did not increase as a result of grassland renovation, but they could have been elevated before March 2009. This is unlikely, as we did not observe any increase in the dissolved organic N concentrations and losses measured from August 2008 (Necpálová *et al.*, 2013). These findings are consistent with the findings of MacDonald *et al.* (2011) who observed reduced DOC fluxes in poorly drained grassland following a single ploughing event as a result of DOC adsorption and thus DOC retention and mineralization. Consequently, the heterotrophic respiration of the soil microbial biomass (mineralization) was considered to be a single potential pathway responsible for the SOC loss from the RG. The decomposition rates are generally controlled by the chemical composition of SOM and edaphic, climatic and hydrological conditions (Post *et al.*, 2001). However, on the basis of available SOC data from four sampling occasions spanning a period of two-and-a-half years, the relationships with the hydrological and climate variables could not be fully established. Despite this, a weak correlation between change in SOC and soil temperature ($P = 0.06$; $R^2 = 0.87$; data not presented) indicated that mineralization rates tended to be enhanced by higher temperature (June to October). The strong dependence of soil respiration following grassland ploughing on soil temperature has been previously reported (Angers *et al.*, 2010).

The C/N ratio did not change following renovation and was within a range for grassland topsoils (Brogan,

1966; Jenkinson, 1988; Ammann *et al.*, 2007). The narrow range of C/N ratio indicated that N dynamics followed SOC dynamics very closely, as has been observed in other PG studies (e.g. Ammann *et al.*, 2009; Wang *et al.*, 2009). Renovation decreased the topsoil total N content by 3.06 t N ha⁻¹ ($P < 0.05$; Necpálová *et al.*, 2013), and caused its 14% reduction, compared with PG, in the 1.7 years following renovation. Hence, the changes in SOC resulted in concomitant changes in soil N. Such high losses following occasional renovation of PG have implications for biochemical processes related to C and N cycling, which may further determine GHG exchange in PG ecosystem.

A recent survey analysis of 453 Irish farms reported that 6.8% of the land area on dairy farms in Ireland is reseeded by ploughing annually (Creighton *et al.*, 2010). Due to recent intensification of grass-based production systems, this proportion is expected to increase, to at least 8–10% (Shalloo *et al.*, 2010). For these reasons, high SOC losses following renovation could have implications for national GHG inventories, and practices preserving SOC should be taken into full consideration.

SOC change over time under permanent grassland

The SOC dynamic under PG was observed over a 7-year period. Although the successive monitoring involved bulking a large number of soil cores ($n = 45$) per replicate to overcome spatial heterogeneity, the SOC content exhibited a high spatial and temporal variation (C.V. = 11%). The spatial heterogeneity could have been associated with the fact that some paddocks were renovated in 1985 and 1995, and SOC levels were still returning to their previous levels. The interannual variation might have been caused by extremely humid weather conditions in 2008 and 2009, followed by extremely dry conditions in 2010. This hampered the possibility of detecting the SOC change using the proposed sampling strategy within the study period. The statistical approach using a simple linear regression indicated a tendency of PG to act as a strong sink for CO₂, although this was not significant at $P = 0.05$. Despite these limitations, it is hypothesized that PG at the Solohead site may act as a strong CO₂ sink, as justified by (i) high clay content (Mestdagh *et al.*, 2006); (ii) high soil-moisture content (Hutchinson *et al.*, 2007; O'Brien *et al.*, 2010) associated with high rainfall events (O'Brien *et al.*, 2010); (iii) high primary production (Jones and Donnelly, 2004); and (iv) a high groundwater table contributing to waterlogging. Measurements extended over a longer period of time are therefore needed to fully investigate its strength as a CO₂ sink.

The statistical tendency also indicated that the annual C inputs to the soil were sufficient to maintain or increase SOC under PG. Although the quality and quantity of C inputs measured at 27 months following renovation did not differ between the grassland treatments, there was no evidence of a recovery of SOC to previous levels in RG in a two-and-a-half-year period following ploughing. This suggests that physical destruction due to ploughing and reseeded influenced the long-term SOC dynamics (i.e. net SOC balance) compared with PG, particularly its ability to return to previous levels in the years following renovation. Rasmussen *et al.* (1998) reported that at 50 years after ploughing, the SOC levels in silt loamy soil reached only 50% of the previous content. Consequently, further research is required to investigate factors other than C inputs that influence the ability of RG to restore previous levels of SOC.

DNDC modelling

The DNDC model has reproduced long- and short-term changes in SOC across a range of soil types, land uses and climates, including grassland soils in Germany and the UK (Li *et al.*, 1994, 1997, 2003; Smith *et al.*, 1997). In the present study, the DNDC predicted changes in SOC over time and among the treatments correctly, although it underestimated (by 78%) the absolute value of SOC losses following PG renovation. This could be a result of default assumptions employed in the model. Some model applications use a default assumption that a tillage event increases decomposition rates for a single month, but deep ploughing of PG can disrupt soil aggregates and increase decomposition rates for more than a month (Del Grosso *et al.*, 2011). In the DNDC, there are several changes taking place in the disturbed soil layers following ploughing: (a) the soil from the surface to 25 cm is completely mixed; residues, microbial biomass, humads and passive humus above the tilling depth are redistributed uniformly over the tilling depth; (b) decomposition rates of residues, microbial biomass and humads are increased by three times; (c) soil moisture decreases by 20%; (d) the denitrifier population decreases by 30%; and (e) 16% of resistant humads are converted into labile humads (Li *et al.*, 1994). These changes primarily reflect the changes in soil structure and aeration due to tillage. As it has been assumed that subsequent rainfall events erase the effects of tillage, the tillage effects last until the fourth subsequent rainfall event, at which point they stop (Li *et al.*, 1994). The year 2008 was an exceptionally wet year, with 25 rainfall events in June (five events over 10 mm of rainfall). This implies that the DNDC might have reduced the effects of ploughing and underestimated

the SOC losses following PG renovation compared with the field observations due to increased intensity and pattern of rainfall events in 2008.

Conclusions

In this study, we evaluated the impact of ploughing and renovation of PG on SOC dynamics in a clay loam soil. The results indicated that grassland renovation significantly decreased topsoil SOC content, resulting in a 22% reduction compared with PG. The major difference in SOC due to grassland destruction developed in the first four months following ploughing, and heterotrophic respiration of soil microbial biomass was considered to be a single potential pathway responsible for the SOC loss.

Although the estimates of C inputs did not differ between the treatments, and a simple linear regression indicated a tendency of PG to accumulate SOC, there was no evidence of SOC recovery to previous levels in RG during the study period. We therefore conclude that occasional grassland renovation on heavy clay loam soils has a significant impact on the long-term SOC dynamics and C balance of this ecosystem with further potential implications for national GHG inventories. Given the increase in the area of PG renovated annually as a result of recent intensification of dairy production systems, further research on mechanisms of SOC loss following PG renovation to support these findings, and also on alternative grassland management practices to preserve SOC in PG soils, is warranted.

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